

Ex No: 1

IMPLEMENT EIGHT QUEENS PROBLEM

Aim:

To Implement Eight Queens Problem using backtracking

Procedure:

Program

```
N = 8 # Size of the chessboard (8x8)
```

```
# Function to print the solution
```

```
def printSolution(board):
```

```
    for row in board:
```

```
        for i in range(N):
```

```
            print("Q" if row[i] else ".", end=" ")
```

```
        print()
```

```
    print() # Add a newline for readability
```

```
# Function to check if a queen can be placed on board[row][col]
```

```
def isSafe(board, row, col):
```

```
    # Check the column
```

```
    for i in range(row):
```

```
        if board[i][col]:
```

```
            return False
```

```
# Check the upper left diagonal
```

```
for i, j in zip(range(row - 1, -1, -1), range(col - 1, -1, -1)):
```

```
    if board[i][j]:
```

```
        return False
```

```
# Check the upper right diagonal
```

```
for i, j in zip(range(row - 1, -1, -1), range(col + 1, N)):
```

```
    if board[i][j]:
```

```
        return False
```

```
return True
```

```
# Function to solve the 8 Queens problem using backtracking
```

```
def solve(board, row):
```

```
    if row == N:
```

```
        printSolution(board)
```

```
        return True # Stop after finding one solution
```

```
for col in range(N):
```

```
    if isSafe(board, row, col):
```

```
        board[row][col] = 1 # Place queen
```

```
        if solve(board, row + 1): # Recur to place next queen
```

```
            return True # Stop after first valid solution
```

```
        board[row][col] = 0 # Backtrack (remove queen)
```

```
return False
```

```
# Main function to initialize the board and start solving the problem
```

```
def eightQueens():
```

```
    board = [[0 for _ in range(N)] for _ in range(N)]
```



```
if solve(board, 0):  
    print("One possible solution is:")  
else:  
    print("Solution does not exist")
```

```
# Calling the function  
eightQueens()
```

Output:

```
One of the possible solutions is:
```

```
. . . . Q . . .  
. . . Q . . . .  
. . . . . Q . .  
Q . . . . . . .  
. . . . . . Q .  
. Q . . . . . .  
. . Q . . . . .  
. . . . . . . Q
```

```
Python 3.11.4 (tags/v3.11.4:d2340ef, Jun 7 2023, 05:45:37) [MSC v.1934 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
```

```
>>>
```

```
= RESTART: C:/Users/ELCOT/Desktop/Tamilnesan/REC/poai 1a.py
```

```
One of the possible solutions is:
```

```
. . . Q . . . .
. Q . . . . .
. . . . . Q . .
. . Q . . . . .
. . . . . Q .
. . . . Q . . .
. . . . . . Q
Q . . . . . . .
```

```
>>> |
```